

## RFM Segmentation & Customer Lifetime Value

|                 |                            |
|-----------------|----------------------------|
| <b>Database</b> | AdventureWorks 2019 (OLTP) |
| <b>Tool</b>     | SQL — Microsoft SQL Server |
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| <b>Class</b>    | BK-DA-PTHB260409           |

*An end-to-end customer segmentation project built entirely in SQL —  
from raw transactional data to actionable business insights.*

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## Foreword

Dear Lecturer,

The goal of this project is to optimize the customer base in order to grow revenue and strengthen customer relationships through investment in customer experience. I chose to analyze the business's loyal customer segment and develop actionable recommendations to maximize this group's value.

To make the analysis practical and clear, I used data-driven modeling to derive concrete metrics — providing a solid foundation for proposing reasons and solutions that can help the business improve customer retention and experience.

Through this project, I hope to deliver recommendations that support data-driven decision-making around customer experience optimization, ultimately making it easier to grow revenue in a more targeted and intuitive way.

Throughout this project, I did not perform all calculations manually. I leveraged AI assistance for computation and initial suggestions, which I then refined and contextualized to fit the business scenario.

Sincerely,

Nguyễn Tài Trung

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## Part 1 — Ad-Hoc Analysis Questions

*Objective: Answer business-relevant analytical questions around two core themes — identifying high-value customer groups and detecting at-risk customers.*

### Issue 1 — Identifying High-Value Customers

| # | Analytical Question                                                                            | Why It Matters                                                                          | Calculation Logic                                                                                                  | Key Insight                                                                                                                                                                        |
|---|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Classify all customers into 4 groups — Loyal, At Risk, Churn, Potential — using the RFM model. | Creates the official segmentation — the foundation for all subsequent analysis.         | Score each R, F, M dimension from 1–4 using NTILE(4). Combine scores, then use CASE WHEN to assign segment labels. | Customers split into 4 RFM segments, enabling the business to understand behavioral patterns and build data-driven retention strategies.                                           |
| 2 | How many customers are in each RFM group and what % of total do they represent?                | Understand group sizes — know exactly how many people you're targeting in each segment. | Count customers per group, divide by total × 100. Use window function SUM() OVER() for the denominator.            | Churn is the largest group at 32.86%. Potential (25.28%) and Loyal (21.38%) show healthy engagement levels. At-risk customers still represent a notable share requiring attention. |
| 3 | Who are the Top 20 customers by total revenue?                                                 | Identify the specific VIP list — who contributes the most to the business.              | Sum TotalDue per CustomerID, sort descending, take top 20 rows.                                                    | Surprisingly, the Churn group accounts for 13% of total revenue among the Top 20 — confirming that some historically high-value customers have already disengaged.                 |
| 4 | What % of total customers is the Loyal group, and what % of total revenue do they contribute?  | Test the Pareto principle — does this small group contribute disproportionately?        | Filter Loyal from the RFM table, sum their revenue, divide by total revenue × 100.                                 | Loyal represents 21.38% of customers but only 0.23% of revenue — reflecting small-ticket, frequent purchase behavior. Highly engaged, but order value needs to be lifted.          |

| # | Analytical Question                                                  | Why It Matters                                                                      | Calculation Logic                                                                   | Key Insight                                                                                                                                                               |
|---|----------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5 | What is the average purchase frequency (F score) of the Loyal group? | Understand the purchase cycle — know when to send reminders to re-engage customers. | Count total orders per Loyal customer, calculate the group average.                 | Loyal records an average F score of ~3.75/4 — the highest across all groups. Confirms strong and consistent engagement with the business.                                 |
| 6 | What is the Average Order Value (AOV) of Loyal vs. other groups?     | Do Loyal customers spend more per order than other groups?                          | Calculate total revenue ÷ order count per RFM group. Compare AOV across all groups. | Loyal AOV = \$71/order — the lowest among all groups. Pattern confirmed: high frequency, low ticket size. Strategy should focus on upselling to increase per-order value. |

## Issue 2 — Detecting and Retaining At-Risk Customers

| # | Analytical Question                                                                   | Why It Matters                                                                            | Calculation Logic                                                               | Key Insight                                                                                                                                                                          |
|---|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7 | What % of customers are Churn, and how much revenue did they historically contribute? | Assess the damage — did the lost group ever have meaningful value?                        | Filter Churn group, count customers and total revenue, calculate % vs. overall. | Churn = \$89.6M in historical revenue and 46.95% of all customers — the largest segment. The business is losing a large, once-valuable cohort, creating serious future revenue risk. |
| 8 | Among At Risk customers, who had the highest historical revenue? (At Risk VIPs)       | Prioritize win-back efforts — this sub-group has the greatest revenue recovery potential. | Filter At Risk, sort by total revenue descending, take top 10.                  | Top 10 At Risk VIPs show revenue ranging from \$2,754 to \$3,221. Despite disengagement signals, these customers represent significant recoverable revenue worth prioritizing.       |

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## Part 2 — Computation Models

*Objective: Explain the logic and formulas of the models used in this project in plain language — no SQL code.*

### Model 1 — RFM Segmentation

RFM is a customer segmentation model based on 3 behavioral dimensions of purchasing:

|   | Letter    | Full Name                                   | How It's Measured                                                                                     |
|---|-----------|---------------------------------------------|-------------------------------------------------------------------------------------------------------|
| R | Recency   | How recently did the customer purchase?     | Smaller R (more recent) → higher score. NTILE(4) sorted ascending: score 4 = most recent buyer.       |
| F | Frequency | Total number of orders placed historically? | Higher F → more purchases → stronger engagement. NTILE(4) sorted descending: score 4 = most frequent. |
| M | Monetary  | Total revenue the customer has contributed? | Higher M → greater financial value. NTILE(4) sorted descending: score 4 = highest spender.            |

After scoring R, F, M (each dimension 1–4), CASE WHEN classifies customers into 4 segments:

| Segment          | Condition                                      | Business Meaning                                                                             |
|------------------|------------------------------------------------|----------------------------------------------------------------------------------------------|
| <b>Loyal</b>     | $R_s \geq 3$ AND $F_s \geq 3$ AND $M_s \geq 3$ | Bought recently, buys often, spends a lot — the golden group. Must be retained at all costs. |
| <b>At Risk</b>   | $R_s \leq 2$ AND $F_s \geq 3$                  | Used to buy frequently but has gone quiet — at risk of churning. Needs early intervention.   |
| <b>Churn</b>     | $R_s \leq 2$ AND $F_s \leq 2$                  | Both recency and frequency are low — nearly lost. Reactivation cost is usually not worth it. |
| <b>Potential</b> | $R_s \geq 3$ AND $M_s \leq 2$                  | Recently re-engaged but hasn't spent much yet — needs incentives to increase order value.    |

## Part 3 — Current Situation & Recommendations

*Note: All figures are drawn from actual query results after running the full analysis pipeline.*

### Current Situation

| Dimension         | Key Figures                                                                                                                                 |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Segmentation      | Loyal: 4,087 customers (21.38%)   At Risk: 3,277 customers (17.14%)   Churn: 6,283 customers (32.86%)   Potential: 4,834 customers (25.28%) |
| Revenue           | Loyal contributes 0.23% of total revenue   AOV Loyal = \$70.70   AOV At Risk = \$613.05   AOV Potential = \$3,835.81                        |
| Retention Signals | Avg F score (Loyal) = 3.75 / 4   Churn total historical revenue = \$89.6M   At Risk VIP highest revenue = \$3,221                           |

- The Loyal group (21.38% of customers) contributes only 0.23% of revenue — small-ticket, frequent behavior (avg F = 3.75/4, AOV = \$70.70/order). Highly engaged but needs upselling.
- The At Risk group (17.14%) had high historical frequency (avg F = 3.00/4) with AOV of \$613.05 — 8× higher than Loyal. Top 10 At Risk VIPs range \$2,754–\$3,221 in revenue. Highest-priority win-back group.
- The Churn group (32.86%) generated \$89.6M historically but now shows avg F = 1.71/4. Behavioral decline is severe — mass reactivation is not cost-effective.
- The Potential group (25.28%) has the highest AOV at \$3,835.81 but unstable frequency. Priority: activate repeat buying to lift both F and M scores.

### Recommended Strategies

| Segment                     | Recommended Strategy                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Loyal (4,087 customers)     | Run a dedicated loyalty program — prioritize points and exclusive perks to reinforce purchase frequency (avg F = 3.75/4). Send periodic reminders based on behavioral cycles to avoid engagement gaps. Introduce higher-value product bundles to lift AOV from \$70.70 — since habits are already established, increasing order size is the natural next step. |
| At Risk (3,277 customers)   | Prioritize reactivation within 30 days — especially the Top 10 At Risk VIPs (\$2,754–\$3,221 in revenue), the group with the greatest recovery potential. Deploy time-limited win-back offers calibrated against group AOV (\$613.05). Tier by historical revenue before rollout — focus resources on the highest-value sub-group first.                       |
| Potential (4,834 customers) | Drive AOV growth via bundles and upselling — high Rs (recent re-engagement) but low Ms. Introduce products frequently purchased by Loyal customers to guide Potential customers toward higher-value habits, gradually increasing their M score.                                                                                                                |

| Segment                 | Recommended Strategy                                                                                                                                                                                                                                                                                |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Churn (6,283 customers) | <p>Serve via automated channels only — do not invest manual resources across all 6,283 Churn customers. Monitor on a 6-month cycle: if a re-engagement signal appears (new order placed), upgrade tier and adjust strategy. Prioritize all active resources toward At Risk and Potential first.</p> |

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## Part 4 — SQL Queries Used

*All queries are organized in execution order. Each block includes a descriptive comment.*

### RFM Core Model — Build #RFM Temp Table

```
-- =====
-- RFM MODEL – MAIN QUERY
-- Creates temp table #RFM with R, F, M scores and segment labels
-- =====

WITH Recency AS (
SELECT DATEDIFF(DAY, MAX(OrderDate), '2014-07-01') AS R,
C.CustomerID AS khach
FROM Sales.SalesOrderHeader AS S
JOIN Sales.Customer AS C ON S.CustomerID = C.CustomerID
GROUP BY C.CustomerID
),
Frequency AS (
SELECT COUNT(H.SalesOrderID) AS F,
C.CustomerID AS khach
FROM Sales.SalesOrderHeader AS H
JOIN Sales.Customer AS C ON H.CustomerID = C.CustomerID
GROUP BY C.CustomerID
),
Monetary AS (
SELECT SUM(H.TotalDue) AS M,
C.CustomerID AS khach
FROM Sales.SalesOrderHeader AS H
JOIN Sales.Customer AS C ON H.CustomerID = C.CustomerID
GROUP BY C.CustomerID
),
Score AS (
SELECT R.khach AS S,
NTILE(4) OVER (ORDER BY R.R ASC) AS Rs,
NTILE(4) OVER (ORDER BY F.F DESC) AS Fs,
NTILE(4) OVER (ORDER BY M.M DESC) AS Ms
FROM Recency AS R
JOIN Frequency AS F ON R.khach = F.khach
JOIN Monetary AS M ON R.khach = M.khach
)
SELECT S AS khach, Rs, Fs, Ms,
CASE
WHEN Rs >= 3 AND Fs >= 3 AND Ms >= 3 THEN 'Loyal'
WHEN Rs <= 2 AND Fs >= 3 THEN 'At Risk'
WHEN Rs <= 2 AND Fs <= 2 THEN 'Churn'
WHEN Rs >= 3 AND Ms <= 2 THEN 'Potential'
ELSE 'Others'
END AS RFM
INTO #RFM FROM Score;
```

### [Q1] RFM Classification — View All Customers

```
SELECT * FROM #RFM;
```

### [Q2] Segment Distribution — Count & Percentage

```
SELECT
RFM,
COUNT(*) AS SoKhach,
CAST(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER() AS DECIMAL(5,2)) AS PhanTram
FROM #RFM
GROUP BY RFM
ORDER BY SoKhach DESC;
```

### [Q3] Top 20 Customers by Total Revenue

```
DECLARE @tongdoanhthu DECIMAL(18,2);
SELECT @tongdoanhthu = SUM(TotalDue) FROM Sales.SalesOrderHeader;

SELECT TOP 20 R.khach, R.RFM, SUM(S.TotalDue) AS tongthu
FROM #RFM AS R
JOIN Sales.SalesOrderHeader AS S ON R.khach = S.CustomerID
GROUP BY R.khach, R.RFM
ORDER BY tongthu DESC;

-- % revenue of Top-20 Churn customers vs. total company
WITH top20 AS (
SELECT TOP 20 R.khach, R.RFM, SUM(S.TotalDue) AS tongthu
FROM #RFM AS R
JOIN Sales.SalesOrderHeader AS S ON R.khach = S.CustomerID
GROUP BY R.khach, R.RFM ORDER BY tongthu DESC
)
SELECT COUNT(*) AS sokhach, SUM(tongthu) AS tongtop20,
CAST(SUM(tongthu) * 100.0 / @tongdoanhthu AS DECIMAL(10,2)) AS phantramdoanhthu
FROM top20 WHERE RFM = 'Churn';
```

### [Q4] Loyal Group — % Customers vs. % Revenue (Pareto Check)

```
DECLARE @totalrev DECIMAL(12,2);
SELECT @totalrev = SUM(TotalDue) FROM Sales.SalesOrderHeader;

WITH doanhthuloyal AS (
SELECT SUM(s.TotalDue) AS totalrev,
SUM(CASE WHEN RFM = 'Loyal' THEN s.TotalDue ELSE 0 END) AS loyal_revenue
FROM #RFM AS r
JOIN Sales.SalesOrderHeader AS s ON r.khach = s.CustomerID
)
SELECT loyal_revenue, totalrev,
CAST(loyal_revenue * 100.0 / totalrev AS DECIMAL(10,2)) AS loyalrev_pct
FROM doanhthuloyal;
```

### [Q5] Average F Score per RFM Segment

```
SELECT RFM,
CAST(AVG(CAST(Fs AS DECIMAL(10,2))) AS DECIMAL(10,2)) AS diemF_trungbinh
```

```

FROM #RFM
GROUP BY RFM
ORDER BY diemF_trungbinh DESC;

```

### [Q6] AOV — Average Order Value by RFM Segment

```

SELECT r.RFM,
COUNT(s.SalesOrderID) AS tongdon,
CAST(SUM(s.TotalDue) AS DECIMAL(18,2)) AS tongdoanhthu,
CAST(SUM(s.TotalDue) / COUNT(s.SalesOrderID) AS DECIMAL(10,2)) AS AOV
FROM #RFM AS r
JOIN Sales.SalesOrderHeader AS s ON r.khach = s.CustomerID
GROUP BY r.RFM
ORDER BY AOV DESC;

```

### [Q7] Churn Group — Customer % & Historical Revenue

```

SELECT r.RFM,
COUNT(DISTINCT r.khach) AS sokhach,
CAST(SUM(s.TotalDue) AS DECIMAL(18,2)) AS tongdoanhthu,
CAST(COUNT(DISTINCT r.khach) * 100.0 / SUM(COUNT(DISTINCT r.khach)) OVER())
AS DECIMAL(5,2)) AS pct_khach
FROM #RFM AS r
JOIN Sales.SalesOrderHeader AS s ON r.khach = s.CustomerID
GROUP BY r.RFM
ORDER BY tongdoanhthu DESC;

```

### [Q8] Top 10 At Risk VIPs by Historical Revenue

```

SELECT TOP 10 r.khach, r.Rs, r.Fs, r.Ms,
CAST(SUM(s.TotalDue) AS DECIMAL(18,2)) AS tongdoanhthu
FROM #RFM AS r
JOIN Sales.SalesOrderHeader AS s ON r.khach = s.CustomerID
WHERE r.RFM = 'At Risk'
GROUP BY r.khach, r.Rs, r.Fs, r.Ms
ORDER BY tongdoanhthu DESC;

```